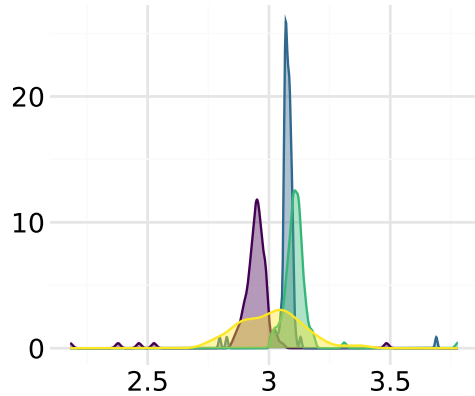
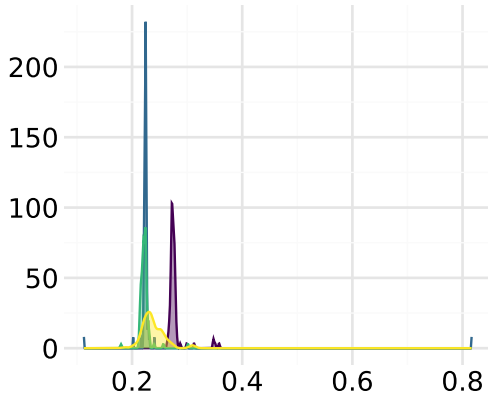
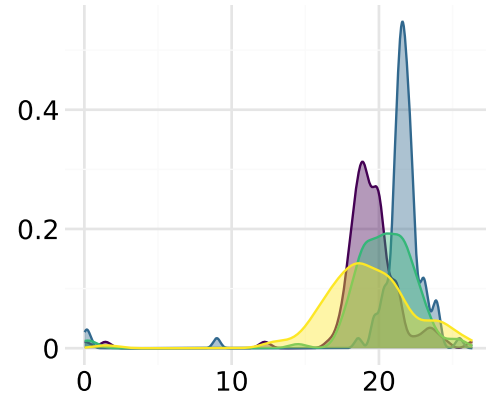
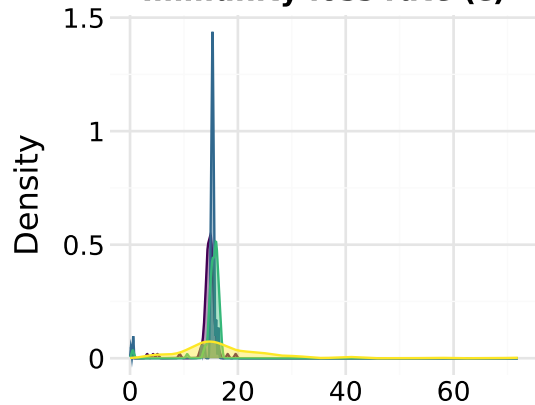
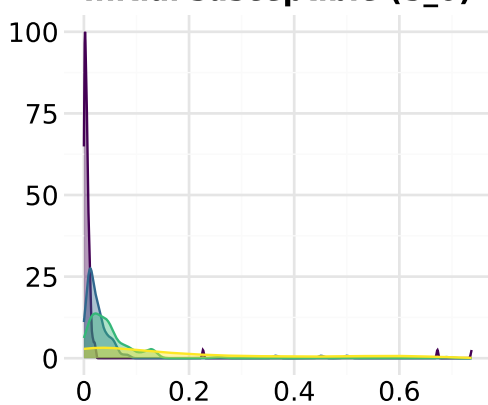
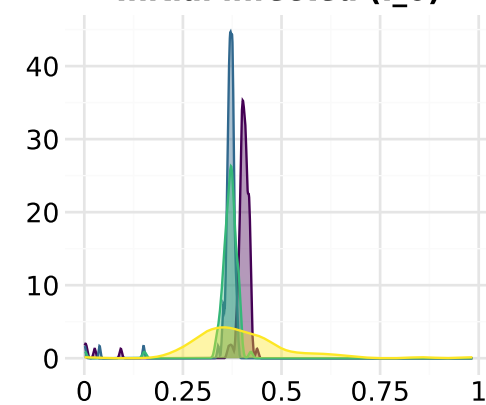
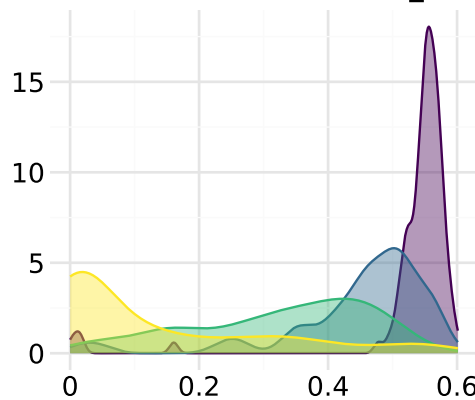
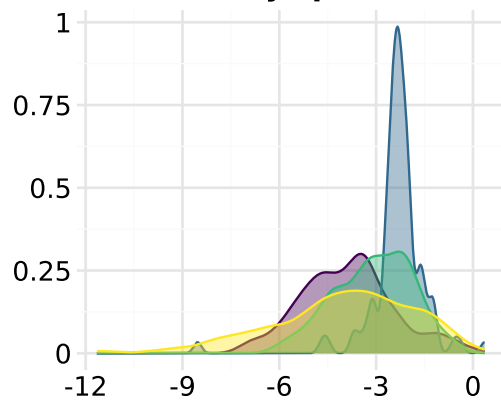
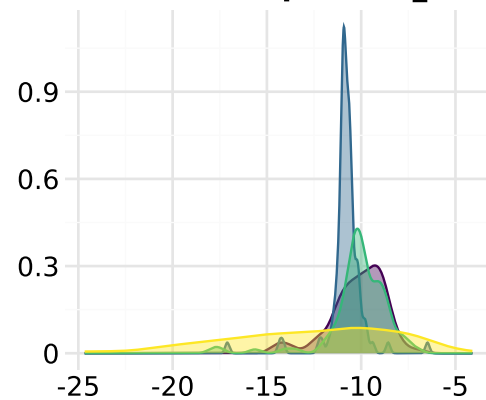


Process noise (σ)**Measurement noise (τ)****Recovery rate (γ)****Immunity loss rate (ϵ)****Initial susceptible (S_0)****Initial infected (I_0)****Initial recovered ($R1_0$)****Seasonality spline (bs3)****Reservoir spline (ω_5)**

Parameter Value Estimate

